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Practical 3

Aim : Create a Cryptocurrency using Python and perform mining in the Blockchain created.

Theory

Blockchain is a distributed digital ledger in which data is stored in blocks connected using cryptographic hashes. Each block contains transactions, timestamp, nonce, previous hash, and its own hash. Mining is the process of finding a valid nonce using Proof of Work. In this experiment, the hash must contain three leading zeros (000). Blockchain validity is checked by verifying the hashes and links between consecutive blocks.

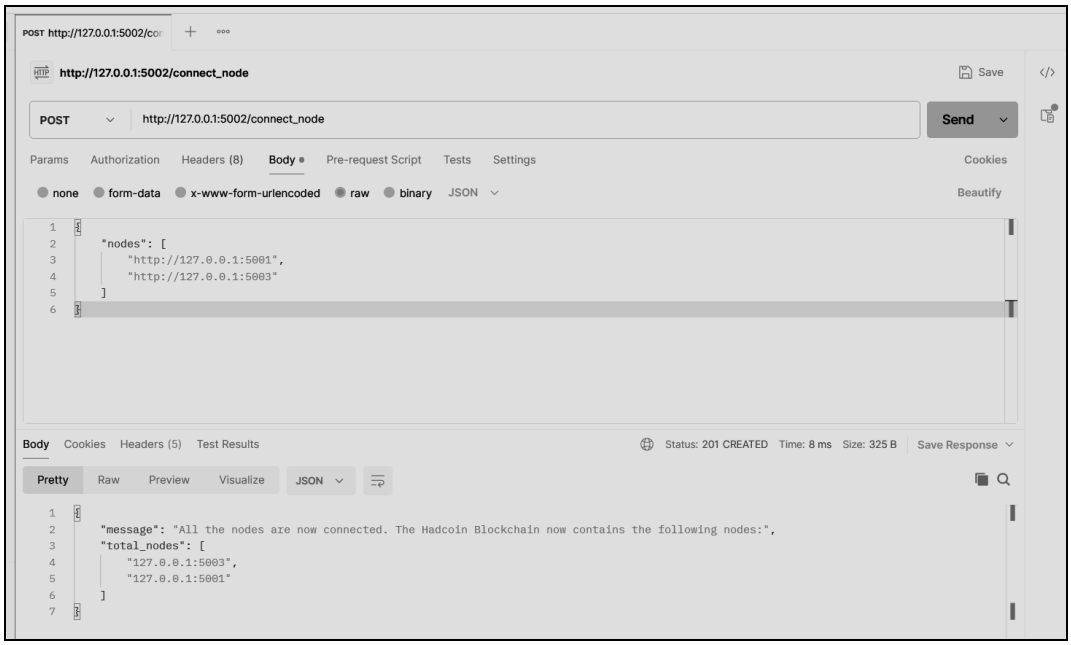
Tasks Performed

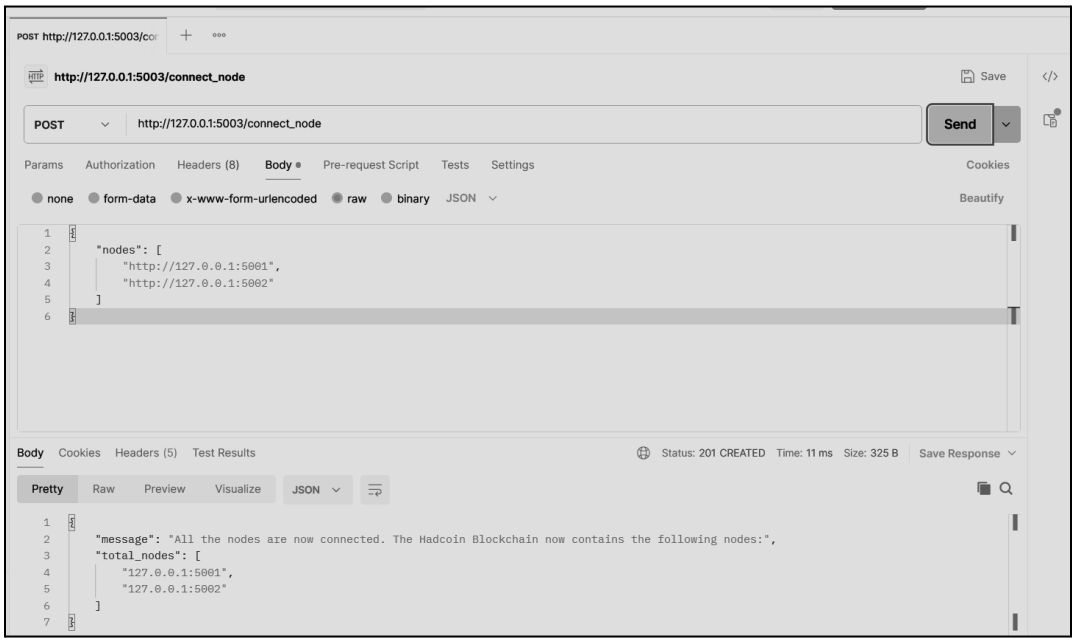
1. Corrected errors in the given Blockchain program.
2. Added the create_transactions() method.
3. Modified mining to occur only when transactions are available.
4. Removed transactions from the pending list after mining.
5. Modified proof_of_work() to find the golden nonce.
6. Implemented the condition of 000 leading zeros in the hash.
7. Tested the Blockchain using Flask and Postman.
8. Displayed the Blockchain and checked its validity.

Program and Output

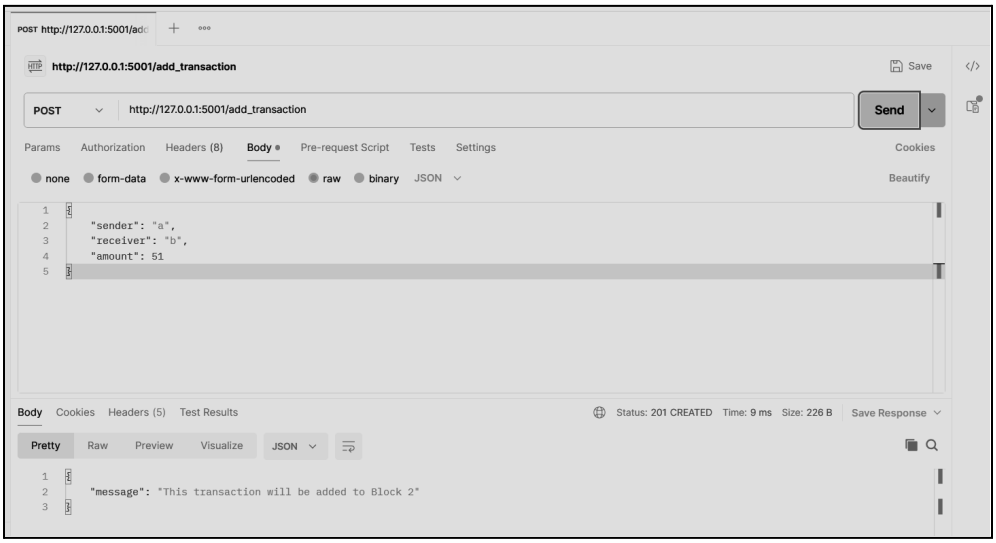
Connecting Nodes Using Postman :

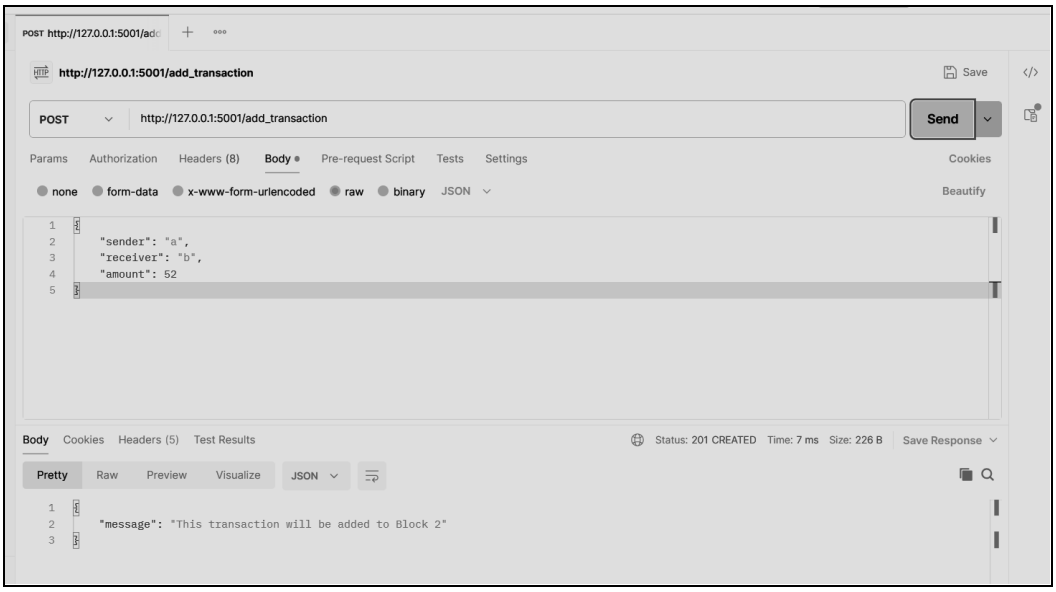


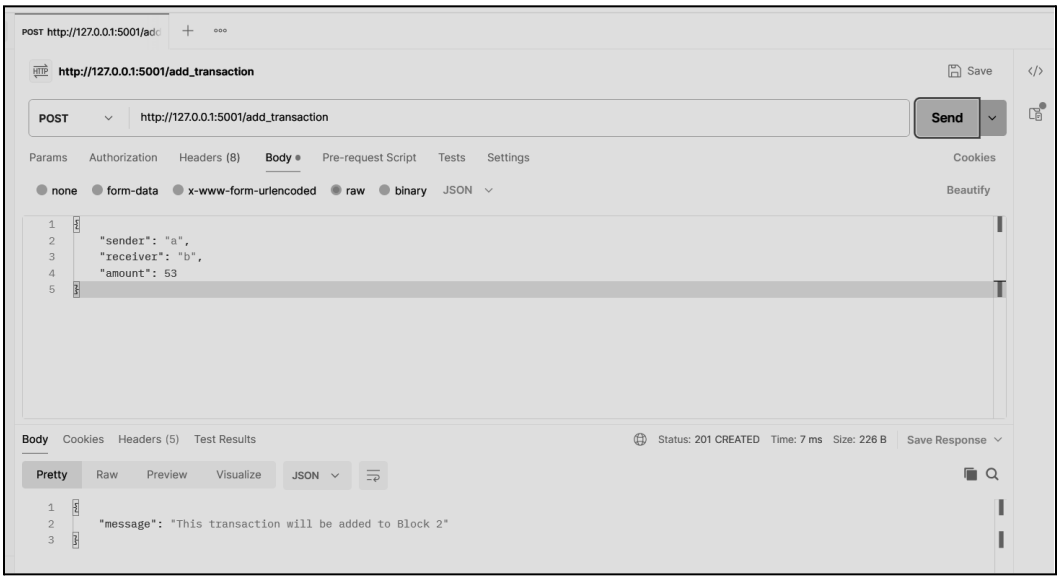




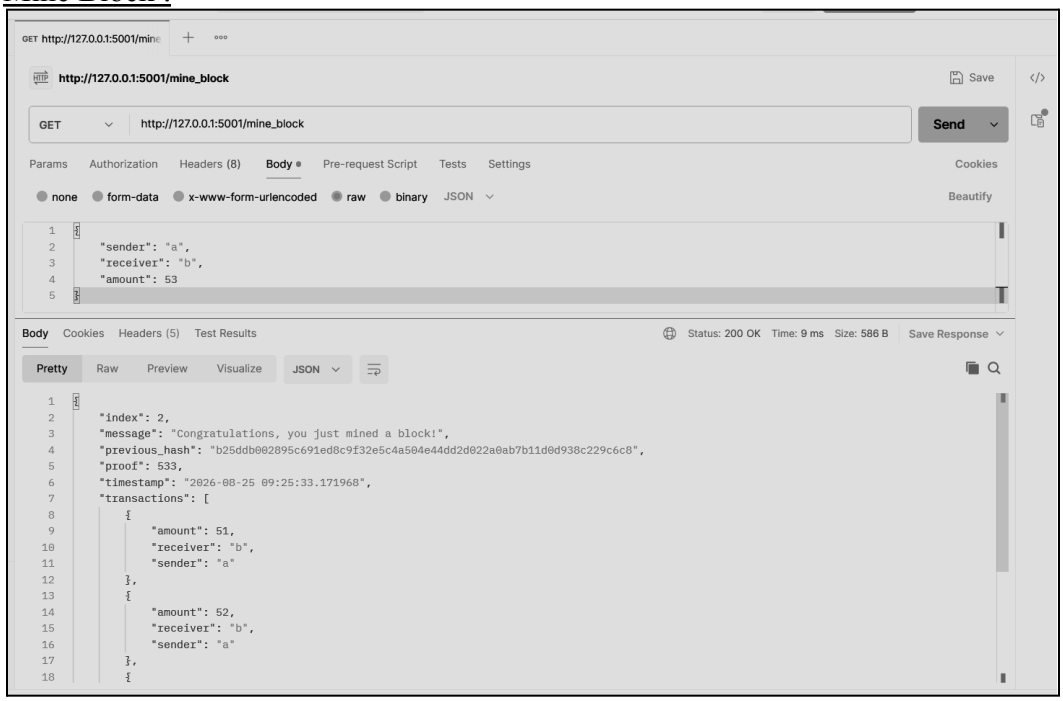
Add 3 transactions :

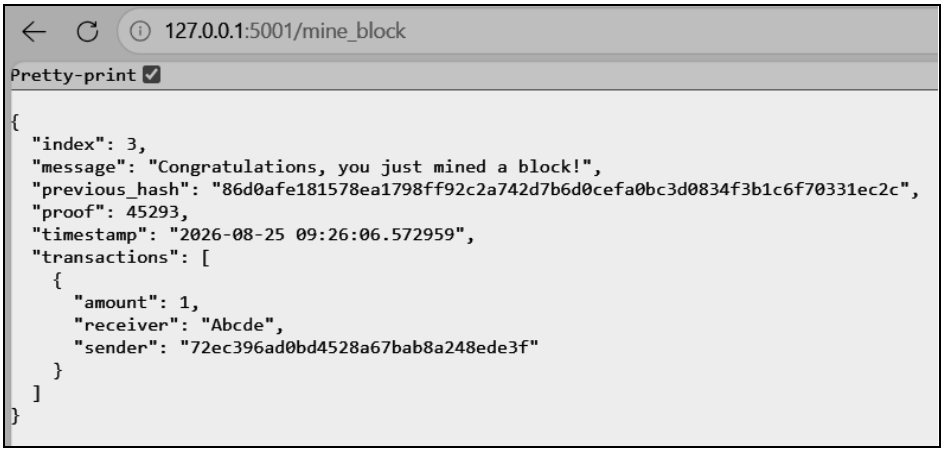



POST http://127.0.0.1:5001/add_transaction
http://127.0.0.1:5001/add_transaction
Save
POST
http://127.0.0.1:5001/add_transaction
Send
Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies
none form-data x-www-form-urlencoded raw binary JSON Beautify
{
"sender": "a",
"receiver": "b",
"amount": 52
}
Body Cookies Headers (5) Test Results
Status: 201 CREATED Time: 7 ms Size: 226 B Save Response
Pretty Raw Preview Visualize JSON
{
"message": "This transaction will be added to Block 2"
}


POST http://127.0.0.1:5001/add_transaction
http://127.0.0.1:5001/add_transaction
Save
POST
http://127.0.0.1:5001/add_transaction
Send
Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies
none form-data x-www-form-urlencoded raw binary JSON Beautify
{
"sender": "a",
"receiver": "b",
"amount": 53
}
Body Cookies Headers (5) Test Results
Status: 201 CREATED Time: 7 ms Size: 226 B Save Response
Pretty Raw Preview Visualize JSON
{
"message": "This transaction will be added to Block 2"
}

Mine Block :


GET http://127.0.0.1:5001/mine_block
http://127.0.0.1:5001/mine_block
Save
GET
http://127.0.0.1:5001/mine_block
Send
Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies
none form-data x-www-form-urlencoded raw binary JSON Beautify
{
"sender": "a",
"receiver": "b",
"amount": 53
}
Body Cookies Headers (5) Test Results
Status: 200 OK Time: 9 ms Size: 586 B Save Response
Pretty Raw Preview Visualize JSON
{
"index": 2,
"message": "Congratulations, you just mined a block!",
"previous_hash": "b25ddb002895c691ed8c9f32e5c4a504e44dd2d022a0ab7b11d0d938c229c6c8",
"proof": 533,
"timestamp": "2026-08-25 09:25:33.171968",
"transactions": [
{
"amount": 51,
"receiver": "b",
"sender": "a"
},
{
"amount": 52,
"receiver": "b",
"sender": "a"
},
{



Check the Chains :





127.0.0.1:5002/get_chain

Pretty-print


```
{
  "chain": [
    {
      "index": 1,
      "previous_hash": "0",
      "proof": 1,
      "timestamp": "2026-08-25 09:18:08.969150",
      "transactions": []
    }
  ],
  "length": 1
}
```


Demonstrate consensus :

127.0.0.1:5002/replace_chain

Pretty-print


```
{
  "actual_chain": [
    {
      "index": 1,
      "previous_hash": "0",
      "proof": 1,
      "timestamp": "2026-08-25 09:17:46.778637",
      "transactions": []
    },
    {
      "index": 2,
      "previous_hash": "b25ddb002895c691ed8c9f32e5c4a504e44dd2d022a0ab7b11d0d938c229c6c8",
      "proof": 533,
      "timestamp": "2026-08-25 09:25:33.171968",
      "transactions": [
        {
          "amount": 51,
          "receiver": "b",
          "sender": "a"
        },
        {
          "amount": 52,
          "receiver": "b",
          "sender": "a"
        },
        {
          "amount": 53,
          "receiver": "b",
          "sender": "a"
        },
        {
          "amount": 1,
          "receiver": "Abcde",
          "sender": "72ec396ad0bd4528a67bab8a248ede3f"
        }
      ]
    },
    {
      "index": 3,
      "previous_hash": "86d0afe181578ea1798ff92c2a742d7b6d0cefa0bc3d0834f3b1c6f70331ec2c",
      "proof": 45293,
      "timestamp": "2026-08-25 09:26:06.572959",
      "transactions": [
        {
```



Validate all 3 blocks :







Conclusion

The Blockchain was successfully implemented using Python. The experiment demonstrated transactions, mining, Proof of Work, golden nonce, hashing, and Blockchain validation. It helped in understanding the basic working and security of Blockchain technology.